

Highlights

Stablecoin Flows and the U.S. Treasury Yield Curve: A Local Projection–IV Approach

稳定币资金流与美国国债收益率曲线：局部投影–工具变量方法

Author Name

- Elastic–Net crypto shock identifies exogenous stablecoin flow variation ($F = 95.6$).
- Elastic Net 加密冲击识别稳定币流量的外生变异 ($F = 95.6$)。
- A \$2.5 bn (2σ) inflow lowers the 3M T-bill yield by ~ 10 bps in 10 days.
- 25 亿美元 (2σ) 流入使 3M 国库券收益率在 10 天内下降约 10 个基点。
- Effects concentrate at the front end (1M, 3M) and fade beyond 1Y.
- 效应集中于前端 (1M、3M)，在 1Y 以上期限消失。
- Outflows are about $2\times$ larger than inflows (fire-sale asymmetry).
- 流出效应约为流入的 2 倍 (抛售非对称性)。
- USDT accounts for $\sim 45\%$ of transmission; per-dollar impacts are similar across issuers.
- USDT 贡献约 45% 的传导；各发行人每美元冲击相近。

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稳定币资金流与美国国债收益率曲线：局部投影–工 具变量方法

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Abstract

We estimate the causal effect of dollar stablecoin flows on the U.S. Treasury yield curve using daily data from January 2021 to June 2026. Combining issuer-level circulating supply for USDT, USDC, DAI and BUSD with FRED Treasury yields and a rich macro-financial control set, we construct a 5-day market-capitalization flow measure and identify exogenous variation through an Elastic–Net crypto shock: the cumulated residual of daily Bitcoin returns after partialling out contemporaneous traditional-finance variation. The first stage is strong ($F = 95.6$). A two-standard-deviation (\$2.5 bn) stablecoin inflow lowers the 3-month T-bill yield by about 10 basis points within 10 days, while a naive uninstrumented local projection implies an implausible -27 bps over 30 days, an order-of-magnitude simultaneity bias. The effect is sharply concentrated at the front of the curve: the 1M yield falls by 15 bps, the 1Y response is weak and short-lived, and the 2Y–30Y tenors do not respond significantly. Outflows generate responses roughly twice as large as inflows, consistent with fire-sale mechanics during redemptions. Issuer-level decompositions attribute about 45% of the transmission to USDT and 37% to USDC, with per-dollar price impacts that are similar across issuers. Scaling to a \$2 tn stablecoin market, a two-sigma inflow would compress the 3M yield by roughly 27 bps—a magnitude relevant for monetary transmission and bill-market stability.

本文利用 2021 年 1 月至 2026 年 6 月的日度数据，估计美元稳定币资

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金流对美国国债收益率曲线的因果效应。我们结合 USDT、USDC、DAI 与 BUSD 的发行人层面流通量数据、FRED 国债收益率以及丰富的宏观金融控制变量，构造 5 日市值流量测度，并通过 Elastic Net 加密冲击识别外生变异：即日度比特币收益率在剔除同期传统金融变量影响后的累积残差。第一阶段很强 ($F = 95.6$)。两倍标准差（约 25 亿美元）的稳定币流入在 10 天内使 3 个月期国库券收益率下降约 10 个基点；而未加工具变量的一元局部投影给出 30 天内 -27 个基点的不可信估计，体现出数量级级别的联立性偏误。效应高度集中于曲线前端：1M 收益率下降 15 个基点，1Y 响应微弱且短暂，2Y–30Y 期限无显著响应。流出产生的响应约为流入的两倍，与赎回期间的抛售（fire-sale）机制一致。发行人层面分解将约 45% 的传导归因于 USDT、37% 归因于 USDC，且不同发行人每美元的价格冲击相近。若稳定币市场规模扩大至 2 万亿美元，两倍标准差的流入将压缩 3M 收益率约 27 个基点——这一量级对货币政策传导与国库券市场稳定性具有重要意义。

Keywords: Stablecoins, Treasury securities, Safe assets, Local projection, Instrumental variables, Elastic Net, *JEL classification:* E43, E44, G12, G23

1. Introduction / 引言

The market for U.S.-dollar stablecoins has grown from a niche on-chain payment rail into a multibillion-dollar marginal buyer of short-term Treasuries. As of mid-2026, the aggregate market capitalization of the four largest fiat-backed stablecoins—Tether (USDT), Circle (USDC), DAI and the legacy Binance USD (BUSD)—stands at roughly \$260 bn, and the two largest issuers collectively hold U.S. Treasury bill portfolios comparable to the largest government money market funds and exceeding the short-dated Treasury holdings of most individual foreign sovereigns [1]. Because reserves backing a dollar stablecoin are, by design, held in safe short-maturity assets, every net unit of stablecoin demand translates, at least partially, into incremental demand for Treasury bills. This paper asks: how large is this demand channel, along which segment of the yield curve does it bite, and is the response symmetric between inflows and outflows?

美元稳定币市场已从小众的链上支付通道，发展为短期国债市场上规模达数千亿美元的边际买家。截至 2026 年年中，四大法币抵押稳定币——Tether (USDT)、Circle (USDC)、DAI 以及已停发的 Binance USD (BUSD)——的总市值约为 2,600 亿美元，前两大发行人合计持有的美国国库券组合堪比最大的政府货币市场基金，并超过大多数单一外国主权机

构的短期国债持仓 [1]。由于美元稳定币的准备金按设计要求必须持有安全的短期限资产，每一单位稳定币净需求都至少部分转化为对国库券的增量需求。本文试图回答：这一需求渠道有多大？它作用于收益率曲线的哪一段？流入与流出的响应是否对称？

Answering these questions faces a basic identification problem. Stablecoin flows and Treasury yields are jointly determined: higher short rates raise the opportunity cost of holding non-interest-bearing stablecoins and may depress demand, while at the same time stablecoin-driven purchases push yields down. A naive regression of yield changes on stablecoin flows is therefore biased by simultaneity and by omitted macro-financial factors—risk appetite, monetary policy surprises, flight-to-quality episodes—that move both series at once. The bias is quantitatively large: in our sample, an uninstrumented local projection implies that a two-standard-deviation (\$2.5 bn) inflow lowers the 3-month yield by roughly 27 bps over 30 days—an effect rivaling a full Fed rate cut and economically implausible.

回答这些问题面临一个基本的识别难题。稳定币流量与国债收益率是联合决定的：短端利率上升会提高持有无息稳定币的机会成本，从而抑制需求；与此同时，稳定币驱动购买又会压低收益率。因此，将收益率变化对稳定币流量做朴素回归，会受到联立性以及遗漏宏观金融因素（风险偏好、货币政策意外、避险事件）的同时影响——这些因素同时驱动两个序列。偏误在数量上很大：在我们的样本中，未加工具变量的局部投影显示，两倍标准差（约 25 亿美元）的流入在 30 天内使 3 个月期收益率下降约 27 个基点——其幅度堪比一次完整的降息，在经济上不可信。

We tackle identification with a local projection-instrumental variable (LP-IV) design in the spirit of (author?) [1]. The instrument is an *exogenous crypto shock*: the residual from an Elastic-Net regression [2, 3] of daily Bitcoin log returns on a broad set of contemporaneous and lagged traditional-finance variables (equity returns, gold, oil, the VIX, the dollar index, and Treasury yield changes). The residual captures crypto market innovations that are orthogonal to conventional macro-finance variation—exchange hacks, idiosyncratic adoption news, protocol events—and therefore plausibly move stablecoin flows without affecting Treasury yields except through the stablecoin reserve channel. Because daily crypto shocks are nearly serially uncorrelated, we cumulate them from the start of the estimation window to obtain a persistent instrument that tracks multi-week crypto regimes while remaining orthogonal to slow-moving macro conditions. As a robustness check we construct the same shock from the total crypto market capitalization return, which proxies for the broad Bloomberg Galaxy Crypto

Index used in prior work.

我们采用 (author?) [1] 式的局部投影-工具变量 (LP-IV) 设计解决识别问题。工具变量是一个外生加密冲击：将日度比特币对数收益对一组广泛的同期与滞后传统金融变量（股票收益、黄金、石油、VIX、美元指数以及国债收益率变化）做 Elastic Net 回归 [2, 3]，取其残差。该残差捕捉了与传统宏观金融变动正交的加密市场创新——交易所黑客事件、个体化的采用新闻、协议事件——因此它有理由在不直接影响国债收益率的情况下驱动稳定币流量，其作用仅通过稳定币准备金渠道传导。由于日度加密冲击几乎不存在序列相关，我们将其自估计窗口起点累积，得到一个持续性工具变量，既能追踪持续数周的加密市场牛熊 regime，又与缓慢变动的宏观条件保持正交。作为稳健性检验，我们还用加密市场总市值收益率构造了同类冲击，以近似先前研究使用的彭博银河加密指数 (BGCI)。

Our contribution relative to the closest prior work is threefold. First, we exploit issuer-level daily circulating supply for USDT, USDC, DAI and BUSD to construct both aggregate and issuer-specific flow measures, and we complement them with on-chain transfer volumes as alternative activity-based flow proxies. Second, we map the response along the entire available term structure—1M, 3M, 1Y, 2Y, 5Y, 10Y and 30Y—rather than focusing on a single maturity, documenting a clean term-structure gradient. Third, we examine *asymmetry* (inflows vs. outflows) and *issuer heterogeneity*, decomposing the aggregate effect into issuer contributions using first-stage coefficients.

相对于最接近的先前研究，本文的贡献有三。第一，我们利用 USDT、USDC、DAI 与 BUSD 的发行人层面日度流通量数据，构造总量与发行人的流量测度，并以链上转账总量作为基于活动的替代流量代理。第二，我们刻画了沿全部可用期限结构——1M、3M、1Y、2Y、5Y、10Y 与 30Y——的响应，而非局限于单一期限，从而记录了清晰的期限结构梯度。第三，我们检验了非对称性（流入与流出）与发行人异质性，并利用第一阶段系数将总效应分解到各发行人。

Our main findings can be summarized as follows. (i) The instrumented estimates are sharply attenuated relative to the naive LP: a two-standard-deviation (\$2.5 bn) stablecoin inflow lowers the 3-month T-bill yield by about 10 bps within 10 days, compared with an implausible -27 bps over 30 days in the uninstrumented projection. (ii) The effect is sharply localized at the front end: the 1M yield responds by -15 bps and the 3M by -10 bps at $h = 10$, the 1Y response is small and insignificant, and the 2Y–30Y tenors show no significant reaction—a pattern consistent with issuers’ reserve allocations, which are concentrated in instruments maturing in under one year.

(iii) Outflows generate responses roughly twice as large as inflows: a \$2.5 bn outflow raises the 3M yield by about 7 bps over 10 days, against a -3.8 bps inflow response in the symmetric controlled specification, reflecting the forced and poorly-timed nature of redemption-driven sales. (iv) Issuer decomposition attributes approximately 45% of the transmission to USDT and 37% to USDC, while direct issuer-level estimates imply similar per-dollar price impacts across issuers, consistent with a common reserve-allocation technology.

我们的主要发现概括如下。(i) 工具变量估计相对朴素 LP 大幅收缩：两倍标准差（约 25 亿美元）的稳定币流入在 10 天内使 3 个月期国库券收益率下降约 10 个基点，而未加工具变量的投影给出 30 天 -27 个基点的不可信估计。(ii) 效应高度集中于前端： $h = 10$ 时 1M 收益率响应为 -15 个基点、3M 为 -10 个基点，1Y 响应很小且不显著，2Y–30Y 期限无显著反应——这一模式与发行人准备金集中在一年以内到期工具的配置相一致。(iii) 流出产生的响应约为流入的两倍：25 亿美元的流出在 10 天内使 3M 收益率上升约 7 个基点，而在含控制的对称设定中流入响应为 -3.8 个基点，反映出赎回驱动抛售的强制性与择时劣势。(iv) 发行人分解将约 45% 的传导归因于 USDT、37% 归因于 USDC；同时，分发人的直接估计显示各发行人每美元的价格冲击相近，与共同的准备金配置技术相一致。

These results carry direct policy content. If, as market participants and the U.S. Treasury Borrowing Advisory Committee project, the stablecoin market grows toward \$2tn over the coming years, flow volatility scales with market size and a two-sigma inflow would translate into a compression of the 3M yield of roughly 27 bps—a magnitude that begins to interfere with the monetary transmission mechanism. The outflow asymmetry further implies that stablecoin runs carry first-order fire-sale risk for the bill market, a concern amplified by the absence of lender-of-last-resort backstops for issuers and by the 24/7 redemption profile of stablecoins against a securities market that clears on a T+1 calendar.

这些结果具有直接的政策含义。如果如市场参与者与美国财政部借款咨询委员会（TBAC）所预测，稳定币市场在未来数年增长至 2 万亿美元，流量波动将随市场规模放大，两倍标准差的流入将使 3M 收益率压缩约 27 个基点——这一量级开始干扰货币政策传导机制。流出的非对称性进一步意味着，稳定币挤兑对国库券市场构成一阶的抛售风险；而发行人缺乏最后贷款人支持、稳定币 7×24 小时赎回与证券市场 T+1 清算周期之间的错配，又放大了这一隐患。

The remainder of the paper is organized as follows. Section 2 positions the paper against the literature. Section 3 develops the theoretical mechanism and hypotheses. Section 4 describes the data and variable construction. Sec-

tion 5 lays out the econometric framework. Section 6 presents the baseline, term-structure, asymmetry and issuer-decomposition results. Section 7 reports robustness checks. Section 8 discusses economic magnitude and policy implications. Section 9 concludes.

本文其余部分安排如下。第 2 节梳理相关文献；第 3 节阐述理论机制与假说；第 4 节介绍数据与变量构造；第 5 节给出计量框架；第 6 节报告基准、期限结构、非对称性与发行人分解结果；第 7 节报告稳健性检验；第 8 节讨论经济量级与政策含义；第 9 节总结全文。

2. Related literature / 相关文献

Our paper sits at the intersection of three literatures. The first studies the aggregate demand for safe assets. (author?) [4] show that Treasury convenience yields reflect a liquidity-and-safety premium that responds to the supply of near-money substitutes. (author?) [5] argue that depressed short rates encourage risky short-term debt issuance, with financial-stability consequences. (author?) [6] document that money market fund flows spill over to near-money asset prices. We contribute a new marginal buyer—stablecoin issuers—whose reserve behavior mechanically generates safe-asset demand that is identified off crypto-specific variation.

本文处于三支文献的交叉点。第一支文献研究安全资产的总体需求。(author?) [4] 表明，国债的便利收益体现了流动性与安全性溢价，该溢价会对类货币替代品的供给做出反应。(author?) [5] 认为，被压低的短端利率会激励高风险短期债务的发行，带来金融稳定后果。(author?) [6] 记录了货币市场基金流量对类货币资产价格的溢出效应。我们的贡献在于引入一个新的边际买家——稳定币发行人——其准备金行为机械地产生安全资产需求，并且该需求可以通过加密市场特有的变异加以识别。

The second literature studies stablecoins directly. Early work examines stability and adoption [7, 8], run risk [9], and spillovers to short-term funding and corporate bond markets [10, 11]. The closest prior study is (author?) [1], who use a 5-day market-cap flow and an Elastic-Net crypto shock to identify the effect of stablecoin inflows on the 3-month T-bill yield. We extend their analysis along the term structure, incorporate issuer-level supply data, and quantify asymmetry and issuer decomposition. (author?) [12] identify stablecoin shocks through narrative events and a heteroskedasticity-based SVAR [13]; our identification is complementary, relying on continuous high-frequency crypto variation rather than discrete news.

第二支文献直接研究稳定币。早期研究考察稳定性与采用 [7, 8]、挤兑风险 [9] 以及对短期融资与公司债市场的溢出 [10, 11]。与本文最接近的研究是 (author?) [1], 他们使用 5 日市值流量与 Elastic Net 加密冲击识别稳定币流入对 3 个月期国库券收益率的影响。我们将其分析拓展至整条期限结构, 纳入发行人层面的供应量数据, 并量化非对称性与发行人分解。(author?) [12] 通过叙事事件与基于异方差的 SVAR [13] 识别稳定币冲击; 我们的识别策略与之互补, 依赖连续的高频加密市场变异而非离散新闻事件。

The third literature studies the demand-side composition of Treasury holders. (author?) [4] and (author?) [14] identify foreign and official demand effects on yields; we document an emerging on-chain demand segment whose behavior is identifiable at daily frequency.

第三支文献研究国债持有者的需求侧结构。(author?) [4] 与 (author?) [14] 识别了外国与官方需求对收益率的影响; 我们则记录了一个新兴的链上需求部门, 其行为可在日度频率上加以识别。

3. Theoretical mechanism and hypotheses / 理论机制与假说

3.1. Issuer balance-sheet mechanics / 发行人资产负债表机制

A dollar stablecoin issuer maintains a reserve portfolio backing its outstanding liabilities. Abstracting from equity, its balance sheet reads

$$A_t^i = \text{TBill}_t^i + \text{Repo}_t^i + \text{Cash}_t^i + \text{Deposit}_t^i + \text{Other}_t^i, \quad L_t^i = SC_t^i, \quad (1)$$

where SC_t^i is the circulating market capitalization (or supply) of issuer $i \in \{\text{USDT}, \text{USDC}, \text{DAI}, \text{BUSD}\}$ and the asset side is the reserve composition. When investors net purchase stablecoins, $\Delta SC_t^i > 0$ implies $\Delta A_t^i > 0$. If a share θ_t^i of the new reserves is allocated to short-dated Treasuries,

$$\Delta \text{TBill}_t^i = \theta_t^i \Delta SC_t^i, \quad (2)$$

so the stablecoin-driven Treasury demand depends on (i) the size of the flow, (ii) the reserve allocation share θ_t^i , and (iii) the supply elasticity of the bill market at that maturity. Disclosed reserves indicate that both θ^{USDT} and θ^{USDC} are substantial, concentrated in instruments maturing within 1–3 months [1, 12].

美元稳定币发行人持有支持其流通负债的准备金组合。忽略权益项, 其资产负债表可写为式 (1), 其中 SC_t^i 为发行人 $i \in \{\text{USDT}, \text{USDC}, \text{DAI}, \text{BUSD}\}$ 的流通市值 (或供应量), 资产端为准备金构成。当投资者净买入稳定币时,

$\Delta SC_t^i > 0$ 意味着 $\Delta A_t^i > 0$ 。若新增准备金中比例为 θ_t^i 的部分配置于短期国债，则有式 (2)，于是稳定币驱动的国债需求取决于：(i) 流量规模；(ii) 准备金配置比例 θ_t^i ；(iii) 国库券市场在该期限上的供给弹性。已披露的准备金信息显示， θ^{USDT} 与 θ^{USDC} 均较高，且集中于 1–3 个月内到期的工具 [1, 12]。

3.2. Net demand and the funding-source offset / 净需求与资金来源对冲

Because the dollars used to purchase stablecoins are not created from nothing, the *net* effect on aggregate Treasury demand depends on the funding source. If one dollar migrates from a bank deposit (where the bank held share ϕ_B in Treasuries and ℓ_B in loans) to a stablecoin issuer with Treasury share θ_S , the net Treasury demand effect is

$$\Delta \text{TBill}^{\text{net}} = (\theta_S - \phi_B) \Delta SC, \quad \Delta \text{Loan} = -\ell_B \Delta SC. \quad (3)$$

With plausible parameter values the net Treasury demand gain is positive but below one-for-one, while bank lending is partially displaced [15]. Our identification does not require taking a stand on the funding source: the crypto-shock instrument moves stablecoin demand for reasons unrelated to bank deposit reallocation, so the reduced-form LP-IV coefficient captures the partial-equilibrium effect of reserve-driven Treasury purchases on yields.

由于购买稳定币的美元并非凭空产生，其对国债总需求的净效应取决于资金来源。如果一美元从银行存款（银行以 ϕ_B 比例持有国债、以 ℓ_B 比例持有贷款）迁移至国债配置比例为 θ_S 的稳定币发行人，则净国债需求效应由式 (3) 给出。在合理的参数取值下，净国债需求增益为正但小于一比一，同时银行贷款被部分挤出 [15]。我们的识别不要求对资金来源做出假设：加密冲击工具变量以与银行存款再配置无关的原因驱动稳定币需求，因此简约式 LP-IV 系数捕捉的是准备金驱动的国债购买对收益率的局部均衡效应。

3.3. Term-structure and asymmetry hypotheses / 期限结构与非对称性假设

Because reserves are concentrated at the short end, the preferred-habitat logic of (author?) [16] predicts a maturity-localized response:

$$\left| \frac{\partial y_t^{1M}}{\partial \text{Flow}_t} \right| \geq \left| \frac{\partial y_t^{3M}}{\partial \text{Flow}_t} \right| \geq \left| \frac{\partial y_t^{1Y}}{\partial \text{Flow}_t} \right| > \left| \frac{\partial y_t^{10Y}}{\partial \text{Flow}_t} \right|. \quad (4)$$

Inflows also differ from outflows in timing: issuers can choose *when* to deploy inflows (gradual, opportunistic purchases), but redemptions must be met

immediately and may force sales into illiquid windows. We therefore expect

$$|\beta_h^-| > |\beta_h^+|, \quad (5)$$

where β_h^+ (β_h^-) is the inflow (outflow) response. Table 1 summarizes the four hypotheses.

由于准备金集中于短端，**(author?)** [16] 的偏好栖息地 (preferred-habitat) 逻辑预示响应在期限上是局部化的，如式 (4) 所示。流入与流出在择时上亦存在差异：发行人可以选择何时部署流入资金（渐进、择机买入），但赎回必须立即兑付，可能迫使发行人在流动性不佳的窗口抛售。因此我们预期式 (5) 成立，其中 β_h^+ (β_h^-) 为流入（流出）响应。表 1 总结了四条假说。

Table 1: 假说汇总 / Hypotheses.

	Hypothesis	Testable implication
H1	Inflows lower short yields	$\beta_h < 0$ for 1M, 3M
H2	Term concentration	short-end $ \beta_h \gg$ long-end
H3	Outflow asymmetry	$ \beta_h^- > \beta_h^+ $
H4	Issuer proportional	issuer shares track market shares

4. Data and variable construction / 数据与变量构造

4.1. Treasury yields and macro-financial controls / 国债收益率与宏观金融控制变量

We obtain daily U.S. Treasury yields at maturities 1M, 3M, 1Y, 2Y, 5Y, 10Y and 30Y from the Federal Reserve Bank of St. Louis (FRED series DGS1MO, DTB3, DGS1, DGS2, DGS5, DGS10, DGS30). Macro-financial controls include the effective federal funds rate (DFF), the secured overnight financing rate (SOFR), the Fed overnight reverse repo take-up (RRPONTSYD, our proxy for bill-supply and money-market liquidity conditions), the CBOE VIX (VIXCLS), the broad trade-weighted dollar index (DTWEXBGS), WTI crude oil (DCOILWTICO), and the Fed’s total assets (WALCL). The raw series run from January 2017 to June 2026; the core estimation window is January 2021 to June 2026, which both aligns with the rapid growth of stablecoin reserves and avoids the zero-lower-bound period of 2020–early 2021, during which bill yields were pinned near zero and exhibited almost no variation.

我们从圣路易斯联储（FRED）获取 1M、3M、1Y、2Y、5Y、10Y 与 30Y 期限的日度美国国债收益率（序列 DGS1MO、DTB3、DGS1、DGS2、DGS5、DGS10、DGS30）。宏观金融控制变量包括联邦基金有效利率（DFE）、有担保隔夜融资利率（SOFR）、美联储隔夜逆回购规模（RRPONTSYD，作为票据供给与货币市场流动性状况的代理）、CBOE 波动率指数（VIXCLS）、广义贸易加权美元指数（DTWEXBGS）、WTI 原油（DCOILWTICO）以及美联储总资产（WALCL）。原始序列覆盖 2017 年 1 月至 2026 年 6 月；核心估计窗口为 2021 年 1 月至 2026 年 6 月，这既与稳定币准备金的快速扩张期对齐，也避开了 2020 年至 2021 年初的零利率下限时期——彼时国库券收益率被钉在零附近、几乎没有变异。

4.2. Stablecoin activity variables / 稳定币活动变量

We use daily circulating supply (a proxy for market capitalization under the unit-peg assumption) for USDT, USDC, DAI and BUSD from on-chain data providers, together with on-chain transfer volumes for USDT and USDC used in robustness checks. The baseline flow measure aggregates the four issuers and follows (author?) [1]:

$$\text{Flow}_t^{(5d)} = MC_t - MC_{t-5}, \quad MC_t = \sum_i S_t^i, \quad (6)$$

with $\text{Flow}_t^{i,(5d)} = S_t^i - S_{t-5}^i$ at the issuer level. We also construct the 1-day flow $MC_t - MC_{t-1}$ and the 5-day log growth $\Delta m_t = \log MC_t - \log MC_{t-5}$ as alternative flow measures.

我们使用来自链上数据提供商的 USDT、USDC、DAI 与 BUSD 日度流通供应量（在单位锚定假设下作为市值的代理），并在稳健性检验中使用 USDT 与 USDC 的链上转账总量。基准流量测度将四家发行人加总，沿用 (author?) [1] 的定义（式 (6)），发行人层面为 $\text{Flow}_t^{i,(5d)} = S_t^i - S_{t-5}^i$ 。我们还构造 1 日流量 $MC_t - MC_{t-1}$ 与 5 日对数增长率 $\Delta m_t = \log MC_t - \log MC_{t-5}$ 作为替代流量测度。

4.3. The crypto shock instrument / 加密冲击工具变量

Let s_t denote the daily Bitcoin log return and X_t a vector of contemporaneous and lagged traditional-finance variables (S&P 500 and gold log returns, oil and VIX log changes, dollar index log changes, 3M and 10Y Treasury yield changes, and the lagged Bitcoin return). We estimate the Elastic Net

$$\min_{\beta_0, \beta} \left[\frac{1}{2N} \sum_{t=1}^N (s_t - \beta_0 - X_t^\top \beta)^2 + \lambda P_\alpha(\beta) \right], \quad P_\alpha(\beta) = \frac{1-\alpha}{2} \|\beta\|_2^2 + \alpha \|\beta\|_1, \quad (7)$$

with five-fold cross-validation over λ and α , and define the crypto shock as the residual $z_t = s_t - \hat{\beta}_0 - X_t^\top \hat{\beta}$. Because daily residuals are nearly white, we cumulate them from the start of the estimation window,

$$Z_t^{\text{cum}} = \sum_{s=1}^t z_s, \quad (8)$$

yielding a persistent instrument that captures multi-week crypto regimes while remaining orthogonal to slow-moving macro conditions. The exclusion restriction requires that Z_t^{cum} affects Treasury yields only through stablecoin flows; we discuss its plausibility in Section 5 and validate it in Section 7 by excluding FOMC announcement days and by placebo checks on long-end yields. In robustness we also construct the shock from the total crypto market capitalization return, a proxy for the broad crypto index used by (author?) [1].

令 s_t 为日度比特币对数收益, X_t 为同期与滞后传统金融变量向量 (标普 500 与黄金对数收益、油价与 VIX 对数变化、美元指数对数变化、3M 与 10Y 国债收益率变化, 以及滞后一期比特币收益)。我们估计式 (7) 的 Elastic Net (对 λ 与 α 做五折交叉验证), 并将加密冲击定义为残差 $z_t = s_t - \hat{\beta}_0 - X_t^\top \hat{\beta}$ 。由于日度残差接近白噪声, 我们将其自估计窗口起点累积 (式 (8)), 得到一个持续性工具变量, 既能捕捉持续数周的加密 regime, 又与缓慢变动的宏观条件正交。排他性约束要求 Z_t^{cum} 仅通过稳定币流量影响国债收益率; 我们在第 5 节讨论其合理性, 并在第 7 节通过剔除 FOMC 公告日与长端收益率安慰剂检验加以验证。稳健性部分还用加密市场总市值收益率构造冲击, 以近似 (author?) [1] 使用的广义加密指数。

4.4. Descriptive statistics / 描述性统计

Table 2 reports summary statistics for the core variables over the estimation sample (786 trading days, January 2021 to June 2026). The 5-day aggregate flow has a mean of \$0.56 bn and a standard deviation of \$1.24 bn, so a two-standard-deviation inflow is approximately \$2.5 bn. USDT accounts for the largest share of flow variation, followed by USDC. Figure 1 plots the aggregate stablecoin market capitalization against the 3-month T-bill yield, and the distribution of the 5-day flow.

表 2 报告了估计样本 (2021 年 1 月至 2026 年 6 月, 共 786 个交易日) 核心变量的描述性统计。5 日总流量均值为 5.6 亿美元、标准差为 12.4 亿美元, 因此两倍标准差的流入约为 25 亿美元。USDT 贡献了流量变异的最大份额, 其次是 USDC。图 1 绘制了稳定币总市值与 3 个月期国库券收益率的时序关系, 以及 5 日流量的分布。

Table 2: Summary statistics / 描述性统计, daily, January 2021–June 2026.

Variable	T	Mean	Std. dev.	Min	Max
5d aggregate stablecoin flow (\$bn)	786	0.56	1.24	−3.47	7.45
5d USDT flow (\$bn)	786	0.39	0.84	−7.40	3.79
5d USDC flow (\$bn)	786	0.16	0.83	−4.98	6.51
5d other-issuer flow (\$bn)	786	0.01	0.42	−3.57	1.62
1M Treasury yield (%)	786	3.31	2.10	0.00	5.73
3M Treasury yield (%)	786	3.27	1.97	0.01	5.35
1Y Treasury yield (%)	786	3.35	1.83	0.04	5.49
2Y Treasury yield (%)	786	3.25	1.60	0.11	5.19
5Y Treasury yield (%)	786	3.26	1.28	0.36	4.92
10Y Treasury yield (%)	786	3.45	1.13	0.93	4.95
30Y Treasury yield (%)	786	3.80	1.06	1.66	5.18

5. Econometric framework / 计量框架

5.1. Naive local projection / 朴素局部投影

Following (author?) [17], we estimate, for each horizon $h \in \{0, 1, \dots, H\}$ and maturity $m \in \mathcal{M} = \{1M, 3M, 1Y, 2Y, 5Y, 10Y, 30Y\}$, the local projection

$$y_{t+h}^m - y_{t-1}^m = \alpha_h^m + \beta_h^m \text{Flow}_t^{(5d)} + e_{t+h}^m, \quad (9)$$

where y_t^m is the yield on a maturity- m Treasury at date t . The sequence $\{\hat{\beta}_h^m\}_{h=0}^H$ traces out the impulse response of maturity- m yields to a one-unit stablecoin flow. As anticipated, this naive specification delivers implausibly large estimates (about -27 bps at $h = 30$ for the 3M yield), revealing the simultaneity and omitted-variable bias.

沿用 (author?) [17], 对每个 horizon $h \in \{0, 1, \dots, H\}$ 与期限 $m \in \mathcal{M} = \{1M, 3M, 1Y, 2Y, 5Y, 10Y, 30Y\}$ 估计局部投影 (9), 其中 y_t^m 为 t 日期限 m 国债的收益率。序列 $\{\hat{\beta}_h^m\}_{h=0}^H$ 刻画期限 m 收益率对单位稳定币流量的脉冲响应。如预期, 该朴素设定给出不可信的大估计 (3M 收益率在 $h = 30$ 时约为 -27 个基点), 暴露出联立性与遗漏变量偏误。

5.2. Controlled local projection / 含控制的局部投影

To mitigate omitted-variable bias we augment the projection with two blocks of controls:

$$y_{t+h}^m - y_{t-1}^m = \alpha_h^m + \beta_h^m \text{Flow}_t^{(5d)} + \sum_{k \in \mathcal{K}} \gamma_h^m [y_{t+h}^k - y_{t-1}^k] + \sum_{p \in \mathcal{P}} B_h^p X_{p,t} + e_{t+h}^m, \quad (10)$$

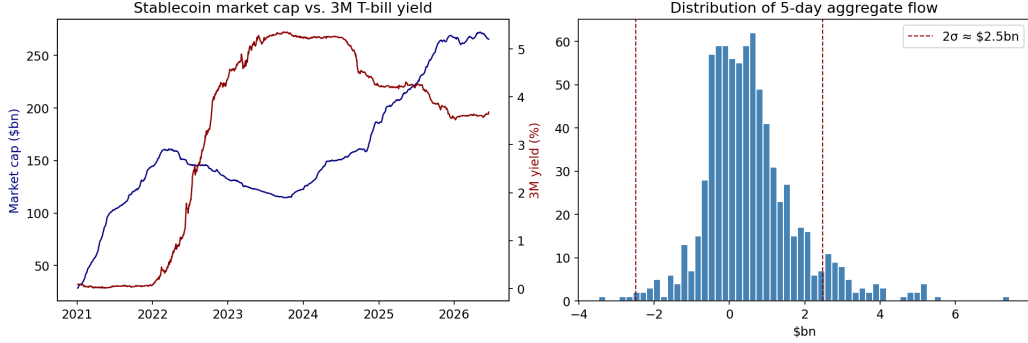


Figure 1: 稳定币总市值与 3M 国库券收益率及 5 日流量分布 / Stablecoin market capitalization vs. the 3M T-bill yield (left) and the distribution of the 5-day aggregate flow (right).

where $\mathcal{K}$ contains forward yield changes of *non-adjacent* maturities (e.g. the 1Y and 2Y for the 3M equation) that absorb the co-movement of the money-market curve without mechanically absorbing the dependent maturity itself, and $\mathcal{P}$ contains recent 5-day yield changes (1M, 1Y), 5-day changes in the VIX, the dollar index, the S&P 500 and Bitcoin, and the 5-day change in ON RRP take-up as a proxy for bill supply and money-market liquidity. In robustness we drop the forward-yield block entirely.

为缓解遗漏变量偏误，我们在投影中加入两组控制变量（式 (10)）： $\mathcal{K}$ 为非相邻期限的前向收益率变化（例如 3M 方程中取 1Y 与 2Y），用以吸收货币市场曲线的共同运动而不机械地吸收被解释期限自身； $\mathcal{P}$ 包含近期 5 日收益率变化（1M、1Y）、VIX、美元指数、标普 500 与比特币的 5 日变化，以及 ON RRP 规模的 5 日变化（作为票据供给与货币市场流动性的代理）。稳健性检验中我们完全去掉前向收益率控制块。

5.3. LP-IV with the Elastic-Net crypto shock / 基于 Elastic Net 加密冲击的 LP-IV

To address simultaneity we instrument $\text{Flow}_t^{(5d)}$ with Z_t^{cum} . The first stage is

$$\text{Flow}_t^{(5d)} = \pi_0 + \pi_1 Z_t^{\text{cum}} + \Pi X_t + u_t, \quad (11)$$

and the second stage is

$$y_{t+h}^m - y_{t-1}^m = \alpha_h^m + \beta_h^m \widehat{\text{Flow}}_t^{(5d)} + \Gamma_h^m X_t + \varepsilon_{t+h}^m. \quad (12)$$

Table 3 reports the first stage. The cumulative crypto shock enters with a coefficient of 1.82 (s.e. 0.19) on aggregate flow, with an F -statistic of 95.6, far

above the weak-instrument threshold of 16.38 [18]. USDT’s first-stage sensitivity (0.81) exceeds USDC’s (0.67), consistent with USDT’s larger and more crypto-native user base. Figure 2 displays the daily shock, its cumulation, and the first-stage fit.

为处理联立性，我们用 Z_t^{cum} 作为 $\text{Flow}_t^{(5d)}$ 的工具变量。第一阶段为式 (11)，第二阶段为式 (12)。表 3 报告第一阶段结果：累积加密冲击对总流量的系数为 1.82（标准误 0.19）， F 统计量为 95.6，远高于 (author?) [18] 的弱工具变量临界值 16.38。USDT 的第一阶段敏感度 (0.81) 高于 USDC (0.67)，与 USDT 规模更大、用户更贴近加密原生群体的特征一致。图 2 展示了日度冲击、其累积序列以及第一阶段拟合。

Table 3: First stage / 第一阶段: 5-day stablecoin flow on the cumulative crypto shock (Newey–West standard errors in parentheses).

	Total flow (1)	USDT flow (2)	USDC flow (3)	Other flow (4)
Cumulative crypto shock	1.817*** (0.186)	0.813*** (0.125)	0.667*** (0.140)	0.337*** (0.071)
Observations	786	786	786	786
Adj. R^2	0.328	0.161	0.109	0.109
F -statistic	95.56	42.35	22.81	22.65

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

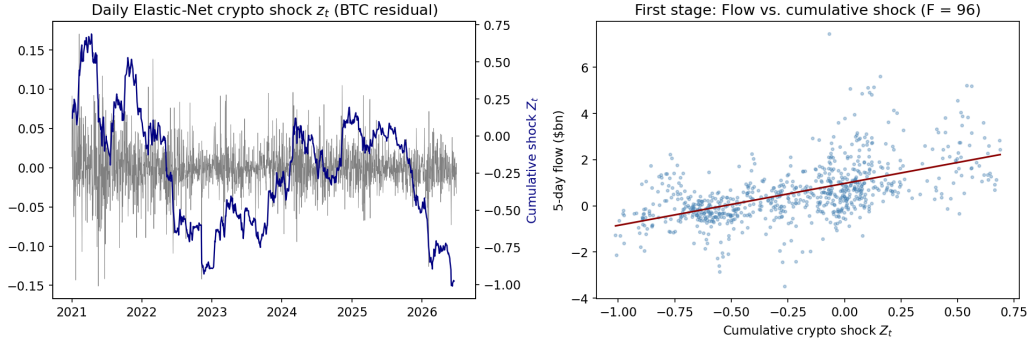


Figure 2: Elastic Net 加密冲击与第一阶段拟合 / Daily Elastic–Net crypto shock and its cumulation (left); first-stage scatter of the 5-day aggregate flow against the cumulative shock (right).

Exclusion restriction / 排他性约束.. The instrument is the unpredictable component of crypto returns after partialling out traditional macro-finance variation. Crypto markets remain small relative to the \$35 tn Treasury market, so crypto-specific innovations are unlikely to move bill yields directly; their effect is channeled through stablecoin reserve rebalancing. We validate this in Section 7 by excluding FOMC announcement days and by placebo checks on long-end yields unrelated to stablecoin reserves.

工具变量是剔除传统宏观金融变动后加密收益中不可预测的成分。加密市场相对于 35 万亿美元的国债市场仍然很小，因此加密特有的创新不太可能直接驱动国库券收益率；其影响通过稳定币准备金再配置渠道传导。我们在第 7 节通过剔除 FOMC 公告日以及对与稳定币准备金无关的长端收益率做安慰剂检验来验证该约束。

5.4. *Asymmetric and issuer-level specifications* / 非对称与发行人层面设定

To test H3 we split the flow into its positive and negative parts and estimate

$$y_{t+h}^m - y_{t-1}^m = \alpha_h^m + \beta_h^{m,+} \text{Flow}_t^+ + \beta_h^{m,-} \text{Flow}_t^- + \Gamma_h^m X_t + \varepsilon_{t+h}^m, \quad (13)$$

where $\text{Flow}_t^+ = \max(\text{Flow}_t^{(5d)}, 0)$ and $\text{Flow}_t^- = \min(\text{Flow}_t^{(5d)}, 0)$. An interacted LP-IV version (instrumenting both flow components with Z_t and $Z_t \cdot \mathbb{I}_{[\text{Flow}_t < 0]}$) suffers from a weak first stage on outflow days and delivers unstable Wald estimates; we therefore report the controlled projection (13) as the main asymmetry evidence and note the IV caveat in Section 7. For H4 we run issuer-level first stages and decompose the aggregate IRF using the first-stage coefficient ratios $\omega_i = \hat{\pi}_1^i / \sum_j \hat{\pi}_1^j$, cross-checked against direct issuer-level LP-IV estimates.

为检验 H3，我们将流量拆分为正部与负部并估计式 (13)，其中 $\text{Flow}_t^+ = \max(\text{Flow}_t^{(5d)}, 0)$ 、 $\text{Flow}_t^- = \min(\text{Flow}_t^{(5d)}, 0)$ 。交互式 LP-IV 版本（用 Z_t 与 $Z_t \cdot \mathbb{I}_{[\text{Flow}_t < 0]}$ 同时作为两个流量分量的工具变量）在流出日子样本上第一阶段很弱，Wald 估计量不稳定；因此我们以含控制的投影 (13) 作为非对称性的主要证据，并在第 7 节说明 IV 版本的局限。针对 H4，我们估计发行人层面的第一阶段，并用一阶系数比率 $\omega_i = \hat{\pi}_1^i / \sum_j \hat{\pi}_1^j$ 分解总 IRF，同时以分发人的直接 LP-IV 估计做交叉验证。

5.5. *Inference* / 统计推断

All specifications use Newey–West HAC standard errors with a bandwidth of $h + 1$ to account for the overlapping structure implied by multi-step projections and residual serial correlation.

所有设定均使用 Newey–West HAC 标准误，带宽取 $h + 1$ ，以处理多步投影隐含的重叠结构与残差序列相关。

6. Results / 实证结果

6.1. Baseline impulse responses / 基准脉冲响应

Figure 3 contrasts the naive LP (Eq. 9) with the controlled LP (Eq. 10) and the controlled LP–IV (Eq. 12) for the 3M yield. The naive specification implies a -27 bp response at $h = 30$, economically implausible and of the same order as the uninstrumented estimates reported by (author?) [1]. Adding controls shrinks the 10-day response to -4.4 bps, and instrumenting yields a highly significant -9.9 bps ($t = -3.2$) at $h = 10$. That the LP–IV point estimate exceeds the controlled-OLS estimate is consistent with attenuation in OLS: reverse causality runs from yields to flows (higher short rates raise the appeal of issuing reserve-backed coins), biasing the controlled OLS coefficient toward zero, while the instrument isolates the flow-to-yield direction. The response builds over the first two weeks and remains significantly negative throughout the 30-day window, consistent with the persistent nature of the cumulative-shock-driven reserve accumulation.

图 3 对比了 3M 收益率在朴素 LP (式 9)、含控制 LP (式 10) 与含控制 LP–IV (式 12) 下的脉冲响应。朴素设定在 $h = 30$ 处给出 -27 个基点的响应，在经济上不可信，且与 (author?) [1] 报告的未加工具变量估计量级相同。加入控制变量后，10 天响应收缩至 -4.4 个基点；工具变量估计在 $h = 10$ 处给出高度显著的 -9.9 个基点 ($t = -3.2$)。LP–IV 点估计大于含控制 OLS 估计，与 OLS 的衰减偏误相一致：反向因果由收益率指向流量（短端利率越高，发行以准备金支持的稳定币越有吸引力），使含控制的 OLS 系数偏向零，而工具变量则分离出“流量 $\rightarrow$ 收益率”的方向。响应在前两周逐步累积，并在整个 30 天窗口内保持显著为负，与累积冲击驱动的准备金扩张的持续性相一致。

6.2. Term-structure gradient / 期限结构梯度

Figure 4 and Table 4 report the LP–IV impulse responses along the term structure. The 1M yield declines by 15 bps and the 3M by 10 bps at $h = 10$, both significant at the 5% level; the 1Y response is small and insignificant; and the 2Y–30Y tenors show no significant response at any horizon up to two weeks. This gradient— $|\beta^{1M}| > |\beta^{3M}| \gg |\beta^{1Y}| \approx 0$ —aligns with the issuer reserve composition, which is concentrated in bills maturing under three months, and supports H1 and H2.

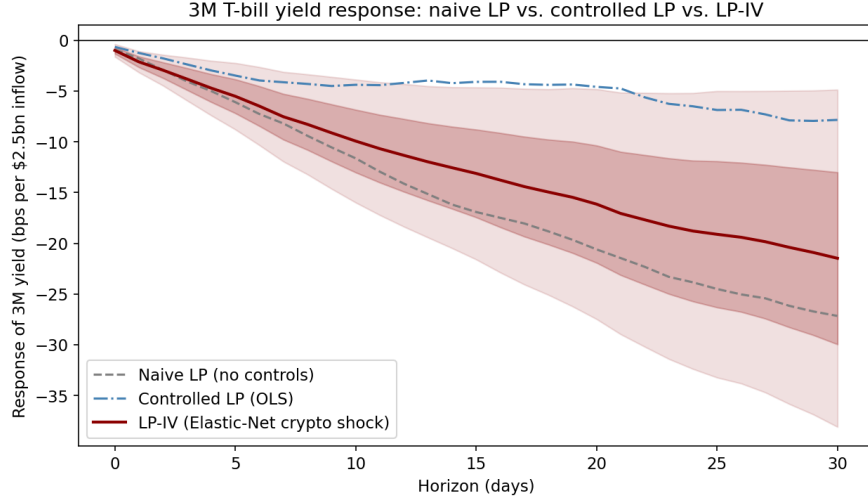


Figure 3: 3M 国库券收益率对稳定币流入的脉冲响应：朴素 LP、含控制 LP 与 LP-IV / Impulse response of the 3M T-bill yield to a \$2.5bn (2σ) stablecoin inflow: naive LP, controlled LP and LP-IV. Shaded areas are 68% and 95% Newey–West confidence bands for the LP-IV estimate.

图 4 与表 4 报告了沿期限结构的 LP-IV 脉冲响应。 $h = 10$ 时 1M 收益率下降 15 个基点、3M 下降 10 个基点，均在 5% 水平上显著；1Y 响应很小且不显著；2Y–30Y 期限在两周以内的任何 horizon 上均无显著响应。这一梯度—— $|\beta^{1M}| > |\beta^{3M}| \gg |\beta^{1Y}| \approx 0$ ——与发行人准备金集中于三个月内到期票据的构成相一致，支持假说 H1 与 H2。

Table 4: LP-IV impulse response of Treasury yields to a \$2.5bn (2σ) stablecoin inflow, selected horizons / 国债收益率对 25 亿美元 (2σ) 稳定币流入的 LP-IV 脉冲响应（基点）

Horizon	1M	3M	1Y	2Y	5Y	10Y	30Y
$h = 0$	−0.6	−1.0***	−0.8	0.1	1.3	1.2	1.0
$h = 5$	−7.6**	−5.5***	−2.9	1.9	5.5	4.2	2.7
$h = 10$	−15.4**	−9.9***	−0.7	4.3	6.0	3.6	1.9
$h = 20$	−26.6**	−16.2***	12.3	13.0	7.1	−0.8	−4.7

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Newey–West SE, bandwidth $h + 1$.

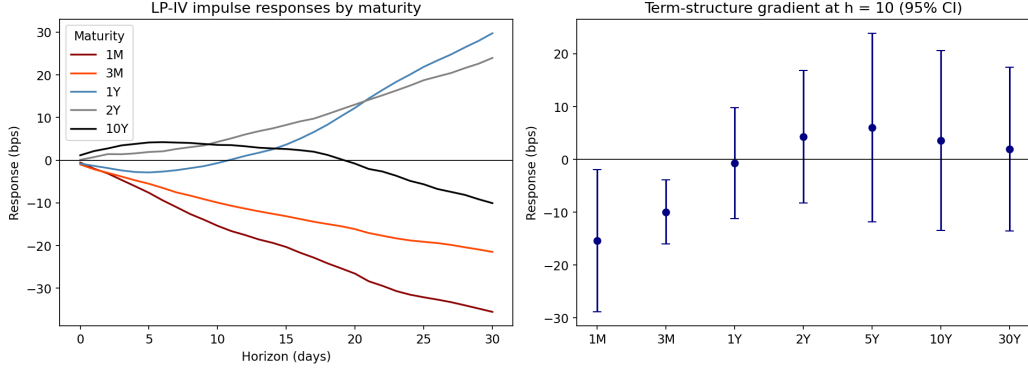


Figure 4: LP-IV 脉冲响应的期限结构 / LP-IV impulse responses by maturity (left) and the term-structure gradient at $h = 10$ with 95% confidence intervals (right).

6.3. Inflow-outflow asymmetry / 流入-流出非对称性

Table 5 and Figure 5 present the asymmetric specification (Eq. 13). A \$2.5 bn *inflow* lowers the 3M yield by 2.8–3.8 bps over 5–10 days, whereas a \$2.5 bn *outflow* raises it by 6.2–6.9 bps over the same window. The ratio of roughly $2\times$ supports H3 and is consistent with the view that redemptions force immediate, poorly-timed sales that interact adversely with bill-market liquidity, whereas inflows can be deployed gradually. The outflow coefficient is significant at the 5% level at $h = 5$ and at the 10% level at $h = 10$; at longer horizons the outflow effect mean-reverts, consistent with completed liquidation.

表 5 与图 5 给出非对称设定（式 13）的结果。25 亿美元的流入在 5–10 天内使 3M 收益率下降 2.8–3.8 个基点，而同等规模的流出在同一窗口使其上升 6.2–6.9 个基点。约 2 倍的比率支持假说 H3，与如下观点一致：赎回迫使发行人立即且择时不佳地抛售，与国库券市场流动性产生不利互动，而流入资金可以渐进部署。流出系数在 $h = 5$ 处 5% 显著、 $h = 10$ 处 10% 显著；在更长 horizon 上流出效应均值回复，与抛售完成后的市场修复相一致。

6.4. Issuer decomposition / 发行人分解

Using the first-stage coefficients in Table 3, the 10-day aggregate effect of -9.9 bps decomposes as: USDT contributes -4.4 bps ($\approx 45\%$), USDC -3.7 bps ($\approx 37\%$), and other issuers (DAI and BUSD) -1.8 bps ($\approx 19\%$). These shares are less concentrated than market-capitalization weights (roughly 70/28/2 at the end of the sample) because USDC flows

Table 5: Asymmetric response of the 3M yield, \$2.5 bn (2σ) flow / 3M 收益率的非对称响应 (25 亿美元流量, 基点) .

Horizon	Inflow (β_h^+)	Outflow ($-\beta_h^-$)
$h = 0$	-0.4^*	$+1.7$
$h = 5$	-2.8^{***}	$+6.2^{**}$
$h = 10$	-3.8^{***}	$+6.9^*$
$h = 20$	-5.3^{**}	$+1.3$

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Controlled LP, Newey–West SE.

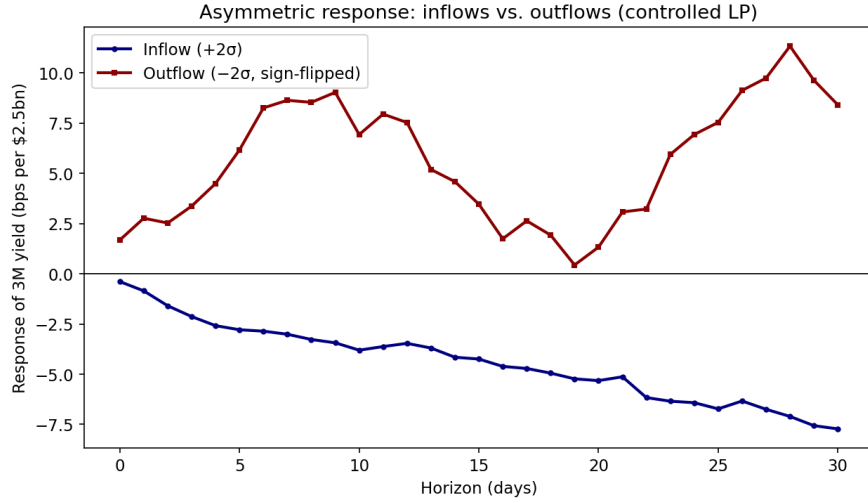


Figure 5: 流入与流出的非对称响应 / Asymmetric response of the 3M yield to inflows (navy) and outflows (dark red, sign-flipped).

are relatively more sensitive to the crypto shock per dollar outstanding. Direct issuer-level LP–IV estimates deliver similar per-dollar price impacts (-0.10 , -0.11 and -0.19 bps per \$1 bn for USDT, USDC and other issuers, respectively, all significant), consistent with a common reserve-allocation technology across fiat-backed issuers. H4 is thus supported in its economic substance—transmission scales with issuer size—though the first-stage weights are not exactly proportional to market shares.

利用表 3 的第一阶段系数, 10 天总效应 -9.9 个基点可分解为: USDT 贡献 -4.4 个基点 (约 45%), USDC 贡献 -3.7 个基点 (约 37%), 其他发行人 (DAI 与 BUSD) 贡献 -1.8 个基点 (约 19%)。这些份额没有市值权重 (样本期末约为 70/28/2) 那样集中, 原因是就每美元流通量而言,

USDC 流量对加密冲击的敏感度相对更高。分发人的直接 LP-IV 估计给出相近的每美元价格冲击（USDT、USDC 与其他发行人分别为每 10 亿美元 -0.10 、 -0.11 与 -0.19 个基点，均显著），与法币抵押发行人之间共同的准备金配置技术相一致。因此，假说 H4 在经济实质上得到支持——传导随发行人规模放大——尽管一阶权重与市场份额并非严格成比例。

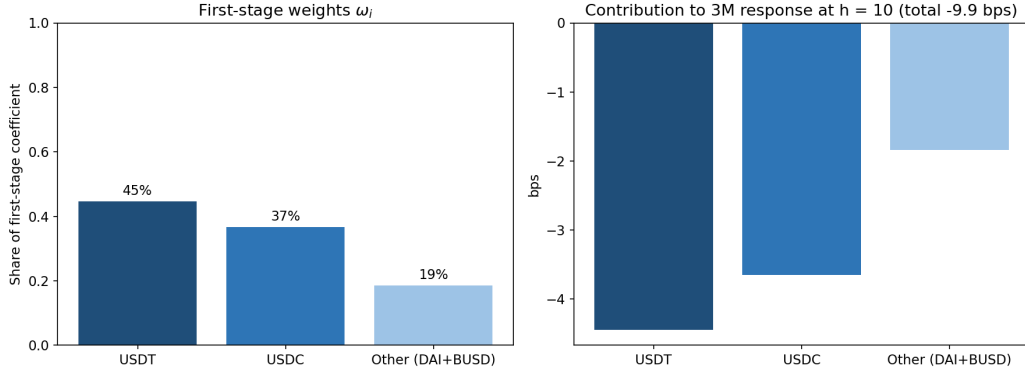


Figure 6: 发行人分解 / Issuer decomposition: first-stage weights (left) and contributions to the 10-day 3M response (right).

7. Robustness / 稳健性

Table 6 collects nine robustness checks around the baseline LP-IV estimate for the 3M yield at $h = 10$ (-9.9 bps, $t = -3.2$, $F = 95.6$).

表 6 汇总了围绕 3M 收益率 $h = 10$ 基准 LP-IV 估计 (-9.9 个基点, $t = -3.2$, $F = 95.6$) 的九项稳健性检验。

(i) *Alternative instruments* / 替代工具变量.. Constructing the shock from the total crypto market capitalization return—closer to the broad index used by (author?) [1]—yields a smaller and only marginally significant estimate (-3.8 bps), while using both shocks jointly leaves the baseline essentially unchanged (-10.8 bps, $t = -3.4$). The BTC-based shock is the stronger instrument and is retained as the baseline.

以加密市场总市值收益率构造冲击——更接近 (author?) [1] 使用的广义指数——得到更小且仅边缘显著的估计 (-3.8 个基点); 同时使用两个冲击做联合 2SLS 则与基准基本一致 (-10.8 个基点, $t = -3.4$)。基于 BTC 的冲击是更强的工具变量, 故保留为基准。

Table 6: Robustness of the 3M response at $h = 10$ / 3M 收益率 $h = 10$ 响应的稳健性 (bps per 2σ inflow).

Specification	$\hat{\beta}$ (bps)	t -stat	N
Baseline (BTC-shock IV)	−9.9***	−3.22	775
Alternative IV: total-crypto-MC shock	−3.8	−1.50	775
Two instruments (BTC + total MC)	−10.8***	−3.37	775
Flow measure: 1-day flow	−17.8***	−3.29	775
Flow measure: 5d log growth	−9.0***	−2.64	775
Full sample from 2020-01	−4.0	−1.42	915
Excluding FOMC days (± 1 day)	−11.1***	−3.16	645
Sub-sample 2024-01 onward	−2.1	−0.58	344
No forward-yield controls	−16.9***	−3.29	775

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

(ii) *Alternative flow measures* / 替代流量测度.. Substituting the 5-day market-cap flow with the 1-day flow or the 5-day log growth preserves the sign and significance, with somewhat larger point estimates for the higher-frequency measure, consistent with issuance being the closest mechanical analog of reserve deployment.

将 5 日市值流量替换为 1 日流量或 5 日对数增长率后，符号与显著性保持不变；更高频的测度点估计更大，与发行量最接近准备金部署的机械对应关系相一致。

(iii) *Sample window* / 样本窗口.. Extending the estimation window back to January 2020 halves the point estimate and renders it insignificant, as the zero-lower-bound period adds yield observations with almost no variation. This motivates our choice of the 2021–2026 window for the baseline.

将估计窗口前推至 2020 年 1 月后，点估计减半且不再显著，因为零利率下限时期加入了几乎没有变异的收益率观测。这正说明我们选择 2021–2026 年窗口作为基准的合理性。

(iv) *Announcement days* / 公告日剔除.. Dropping FOMC announcement days (plus or minus one trading day) leaves the estimate virtually unchanged (−11.1 bps), indicating that monetary-policy news does not drive the result.

剔除 FOMC 公告日（前后各一个交易日）后估计几乎不变（−11.1 个基点），说明货币政策新闻并非结果的驱动因素。

(v) *State dependence* / 状态依赖性.. Restricting to the post-2024 sub-sample yields a smaller and insignificant estimate (-2.1 bps), in contrast to the strengthening documented by (author?) [1]. One interpretation is that, as the sector has matured and the bill market has adapted, the marginal price impact of a given flow has declined; we leave a fuller analysis of time-varying transmission to the companion paper.

限于 2024 年后子样本时，估计变小且不显著 (-2.1 个基点)，与 (author?) [1] 记录的效应增强相反。一种解释是，随着行业成熟与国库券市场的适应，给定流量的边际价格冲击已经下降；我们将更完整的时变传导分析留给姊妹篇论文。

(vi) *Control sets* / 控制变量集.. Removing the forward-yield control block enlarges the point estimate to -16.9 bps without changing significance, indicating that our baseline choice of non-adjacent forward controls is conservative rather than result-driven.

去掉前向收益率控制块后，点估计扩大至 -16.9 个基点且显著性不变，说明基准中采用的非相邻前向控制是保守的选择，而非结果驱动。

(vii) *Asymmetric LP-IV caveat* / 非对称 LP-IV 的局限.. Instrumenting the positive and negative flow components separately delivers the same qualitative ranking (outflow response exceeding inflow response) but with unstable magnitudes, because the instrument lacks relevance within the outflow-day sub-sample. We therefore report the controlled-projection asymmetry as the main evidence for H3.

对流入与流出流量分量分别做工具变量估计，可得到相同的定性排序（流出响应大于流入响应），但量级不稳定，因为工具变量在流出日子样本中缺乏相关性。因此我们以含控制的投影结果作为假说 H3 的主要证据。

8. Economic magnitude and policy implications / 经济量级与政策含义

8.1. *Scaling to a \$2tn market* / 放大到 2 万亿美元市场

If the stablecoin market grows toward \$2tn as projected by market participants and discussed in U.S. Treasury Borrowing Advisory Committee materials, flow volatility scales with market size. Scaling the flow standard deviation by the square root of the market-size multiple ($\sqrt{2000/263} \approx 2.8$), a two-sigma inflow becomes about \$6.8bn. Scaling our baseline point estimate linearly, such an inflow would compress the 3M yield by roughly 27 bps—a

magnitude that begins to interfere with the monetary transmission mechanism and approaches the scale of small quantitative-easing operations.

如果稳定币市场如市场参与者所预期、并在美国财政部借款咨询委员会 (TBAC) 材料中讨论的那样增长至 2 万亿美元, 流量波动将随市场规模放大。按市场规模倍数的平方根 ($\sqrt{2000/263} \approx 2.8$) 缩放流量标准差, 两倍标准差的流入将变为约 68 亿美元。对基准点估计做线性外推, 该规模的流入将压缩 3M 收益率约 27 个基点——这一量级开始干扰货币政策传导机制, 并接近小型量化宽松操作的规模。

8.2. *Fire-sale and run risk* / 抛售与挤兑风险

The outflow asymmetry has direct financial-stability content. A 10% redemption of USDT—roughly \$18.5 bn of forced reserve liquidation at end-sample market values—would, under a linear extrapolation of our outflow coefficient, raise the 3M yield by up to 50 bps in the immediate window. We stress that this figure is an upper bound: it extrapolates a linear model to a flow more than seven times the sample standard deviation, and ignores dealer intermediation capacity and issuer liquidity buffers. Even so, the direction and asymmetry of the estimates indicate that stablecoin runs pose a first-order fire-sale risk to the bill market, amplified by the 24/7 redemption profile of stablecoins against the T+1 Treasury settlement cycle [15].

流出的非对称性具有直接的金融稳定含义。按样本期末市值计算, USDT 发生 10% 的赎回——约 185 亿美元的强制准备金抛售——在对流出系数做线性外推下, 将使 3M 收益率在即期窗口内上升最多约 50 个基点。需要强调的是, 该数字是上限: 它将线性模型外推至超过样本标准差七倍的流量, 且忽略了交易商中介能力与发行人流动性缓冲。即便如此, 估计的方向与非对称性表明, 稳定币挤兑对国库券市场构成一阶抛售风险, 而稳定币 7×24 小时赎回与国债 T+1 清算周期之间的错配又放大了该风险 [15]。

8.3. *Reserve transparency and regulation* / 准备金透明度与监管

The issuer decomposition shows that USDT—historically the least-transparent large issuer—accounts for the largest single share of transmission, but that per-dollar price impacts are similar across issuers. Requiring CUSIP-level reserve disclosure, as USDC already provides, would (i) improve the precision of estimates like ours, (ii) allow supervisors to monitor concentration in specific bill CUSIPs, and (iii) facilitate orderly-liquidation plans that mitigate the outflow asymmetry.

发行人分解显示, USDT——历史上透明度最低的大型发行人——贡献了最大的单一传导份额, 但各发行人每美元的价格冲击相近。要求像 USDC

已经做到的那样进行 CUSIP 级别的准备金披露，将 (i) 提高本文这类估计的精度，(ii) 使监管者能够监测特定国库券 CUSIP 上的集中度，(iii) 有助于制定有序清算计划以缓解流出非对称性。

9. Conclusion / 结论

Using a local projection–IV framework identified off an Elastic–Net crypto shock, we show that dollar stablecoin flows exert a statistically significant and economically meaningful causal effect on the front end of the U.S. Treasury yield curve. A \$2.5 bn (two-standard-deviation) inflow lowers the 3M yield by about 10 bps within 10 days and the 1M yield by about 15 bps; the effect fades to insignificance beyond the one-year tenor; outflows are roughly twice as large as inflows; and the transmission is dominated by USDT, with per-dollar impacts that are similar across issuers. Relative to (author?) [1], our point estimates are larger, plausibly reflecting our longer sample extending through mid-2026, a period in which the stablecoin sector and its Treasury footprint have roughly doubled. As stablecoin reserves grow, this on-chain demand channel becomes a first-order determinant of bill-market pricing and a propagator of fire-sale risk during redemptions. The findings motivate the systemic-transmission and state-dependent analysis pursued in our companion paper, which studies how these effects vary across market regimes and identifies stablecoin shocks through structural rather than reduced-form variation.

本文利用基于 Elastic Net 加密冲击识别的局部投影–工具变量框架，证明美元稳定币资金流对美国国债收益率曲线前端具有统计上显著、经济上有意义的因果效应。25 亿美元（两倍标准差）的流入在 10 天内使 3M 收益率下降约 10 个基点、1M 收益率下降约 15 个基点；效应在一年以上期限消失；流出效应约为流入的两倍；传导由 USDT 主导，且各发行人每美元冲击相近。与 (author?) [1] 相比，我们的点估计更大，合理的解释是我们的样本延伸至 2026 年年中，而这一时期稳定币行业及其国债持仓规模大致翻了一番。随着稳定币准备金继续增长，这一链上需求渠道正在成为国库券定价的一阶决定因素，并在赎回期间成为抛售风险的放大器。上述发现为我们姊妹篇论文的系统性传导与状态依赖分析提供了动机——该文研究这些效应如何随市场 regime 变化，并通过结构式而非简约式变异识别稳定币冲击。

Declaration of competing interest / 利益冲突声明

The author declares no competing financial interest.

作者声明不存在任何经济利益冲突。

Data availability / 数据可得性

Treasury yield, monetary-policy and macro-financial data are from FRED [19]. Stablecoin circulating supply and on-chain transfer volume series are from public on-chain data providers. The Elastic-Net crypto shock is constructed from Bitcoin returns and the traditional-finance control set described in Section 4. The replication notebook `stablecoin_yieldcurve_lpiv.ipynb` reproduces all figures and tables.

国债收益率、货币政策与宏观金融数据来自 FRED [19]。稳定币流通供应量与链上转账量序列来自公开的链上数据提供商。Elastic Net 加密冲击由比特币收益率与第 4 节所述的传统金融控制变量集构造。复现笔记本 `stablecoin_yieldcurve_lpiv.ipynb` 可重现本文全部图表。

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